

LifeTrac: the game

A farm-machine tycoon whose rule book is a theorem



The last frame of Ada's season, drawn from the position the rule book actually reached: 18 449 in cash, 40 litres in the tank, day 17.125, 20 hectares worked, a LifeTrac in the shed, 28 060 net worth.

Build the tractor, then work the land

You start with 15 000 units of cash, an empty yard and a shelf with nothing on it. The market sells twenty-four raw materials — square tube, plate, bolts, hose, a gear pump, four wheel motors, an engine, a seat, a tin of paint — and the six sub-assembly kits that make up a LifeTrac. The workbench turns a shelf full of materials into a machine. The field planner puts the machine to work: seed and fuel go in, a wheat harvest comes out, and the calendar moves on.

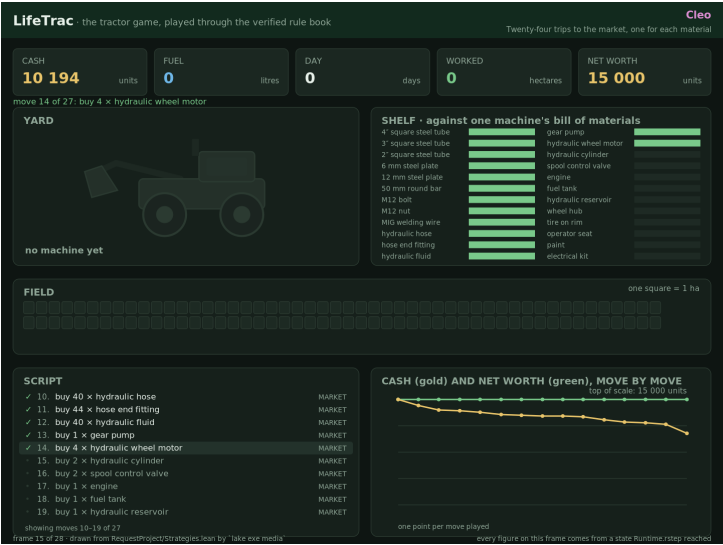
Every price, every bill of materials, every hour of shop time and every litre per hectare in the game is taken from the engineering model of the real machine, formalized in Lean 4: the hydrostatic drive that sets the working speed, the loader that sets the capacity, the production chain that says what a tractor is made of, and the field arithmetic that says how long a hectare takes.

The rule book is proved

The game is not a simulation with a rule book bolted on. The rule book *is* the formalization:

- a move that is not affordable or not possible simply has no result — the type says so;
- cash never goes negative, on any play;
- ordering a kit is net-worth neutral;
- the browser build is compiled from the same Lean rule book, and the bytes it ships are proved to parse back to the module the proofs are about.

What you are looking at



Cleo, halfway through her shopping: move 14 of 27. Nothing built yet, the shelf half full, the net worth flat — buying at the list price moves value from one column to another and no further.

THE HEAD-UP DISPLAY

Cash, fuel, calendar, hectares worked, and net worth: cash plus the shelf at cost plus the fuel plus everything built. It is the figure the compiled WebAssembly module computes, and it is proved to be the net worth of the formal game position.

THE YARD

Empty until the workbench has produced a LifeTrac; then the machine stands in it, and only then will the field planner take an order.

THE SHELF

Twenty-four bars, each filled against what one whole machine needs. Ordering the kit fills them all at once; buying materials one at a time fills them one at a time; building the machine empties them.

THE FIELD

A hundred squares. Each square is a hectare — or, in a play big enough to need it, as many hectares as the caption says.

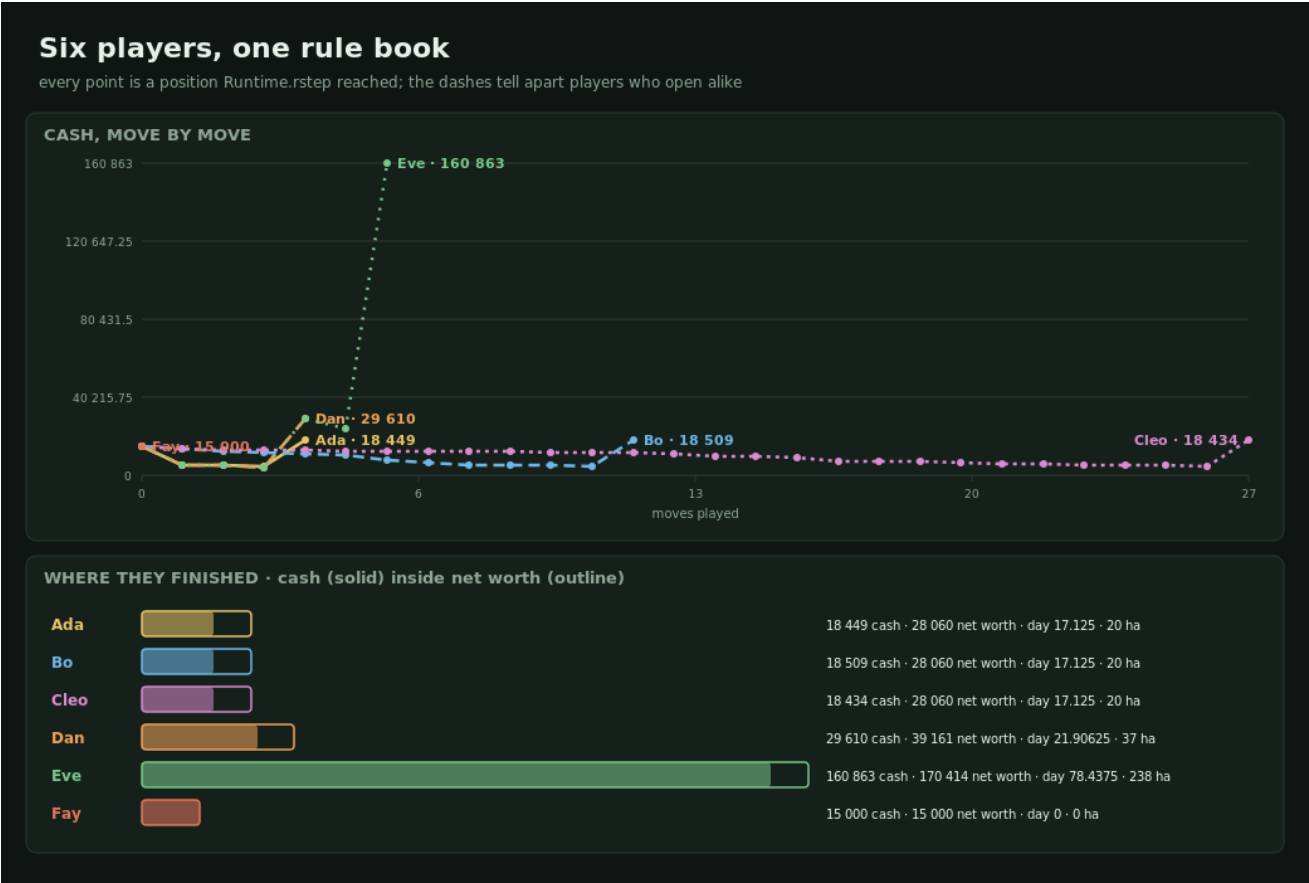
THE SCRIPT AND THE CHART

The moves of the play, ticked off as the rule book accepts them, and the cash and net worth of every position so far.

Six players, one rule book

Six scripts were played through the rule book. Five of them are legal from the first move to the last; the sixth is refused at her first click. The table is the finish, exactly as the game reports it.

Player	moves	cash	net worth	day	hectares
Ada	4/4	18 449	28 060	17.125	20
Bo	12/12	18 509	28 060	17.125	20
Cleo	27/27	18 434	28 060	17.125	20
Dan	4/4	29 610	39 161	21.90625	37
Eve	6/6	160 863	170 414	78.4375	238
Fay	0/2	15 000	15 000	0	0



Cash after every move. Ada, Dan and Eve open identically — order the kit, build it — so their curves lie on top of one another until the fuel is bought; the dashes tell them apart.

The moves, and what the rule book asks of them

- Buy** q units of a material — q whole units, and the cash to pay the list price for them.
- Order** a kit — the cash for the whole bill of materials; everything in it lands on the shelf at once.
- Sell** q units back — the units must be on the shelf, and the yard gets the salvage price, not the list price.

- Build** an assembly — every material of its bill of materials on the shelf; they are consumed, the assembly appears in the yard, and the calendar moves on by the shop hours.
- Refuel** l litres — the cash for them.
- Farm** a hectares — a finished LifeTrac in the yard, the fuel for the work in the tank and the cash for the seed; the harvest comes back the same move.

A move that fails any of these has no result at all: the game’s step function returns nothing, the position does not change, and the button is greyed out on the screen. That is the whole of the rule enforcement — there is no other check anywhere in the game.

The players

Ada

4/4 moves

By the book: one kit, one machine, twenty hectares

cash 18 449 net worth 28 060

day 17.125 20 hectares worked

Cleo

27/27 moves

Twenty-four trips to the market, one for each material

cash 18 434 net worth 28 060

day 17.125 20 hectares worked

Eve

6/6 moves

Plough the harvest back in: 37 ha, then 201 ha

cash 160 863 net worth 170 414

day 78.4375 238 hectares worked

Bo

12/12 moves

Kit by kit, and only the fuel the crop needs

cash 18 509 net worth 28 060

day 17.125 20 hectares worked

Dan

4/4 moves

Every last unit of cash into the first crop: 37 ha

cash 29 610 net worth 39 161

day 21.90625 37 hectares worked

Fay

0/2 moves

Farming before there is a tractor — the rules say no

cash 15 000 net worth 15 000

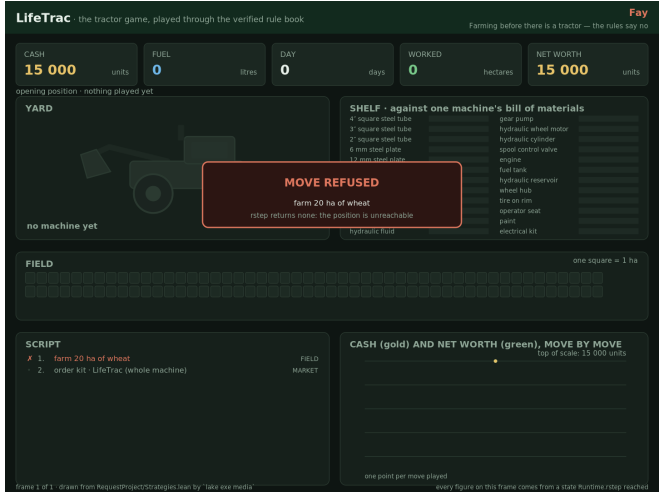
day 0 0 hectares worked

THREE WAYS TO BUY THE SAME MACHINE

Ada orders the whole bill of materials. Bo orders the six kits — frame, four wheel modules, power unit, control station, loader, finishing — and it is proved that this puts exactly the same materials on the shelf for exactly the same money. Cleo goes to the market twenty-four times and pays *fifteen more*: the machine needs one and a half round bars and the shop sells whole ones. All three have a tractor on day 11.5 and finish the season on day 17.125 with 20 hectares behind them.

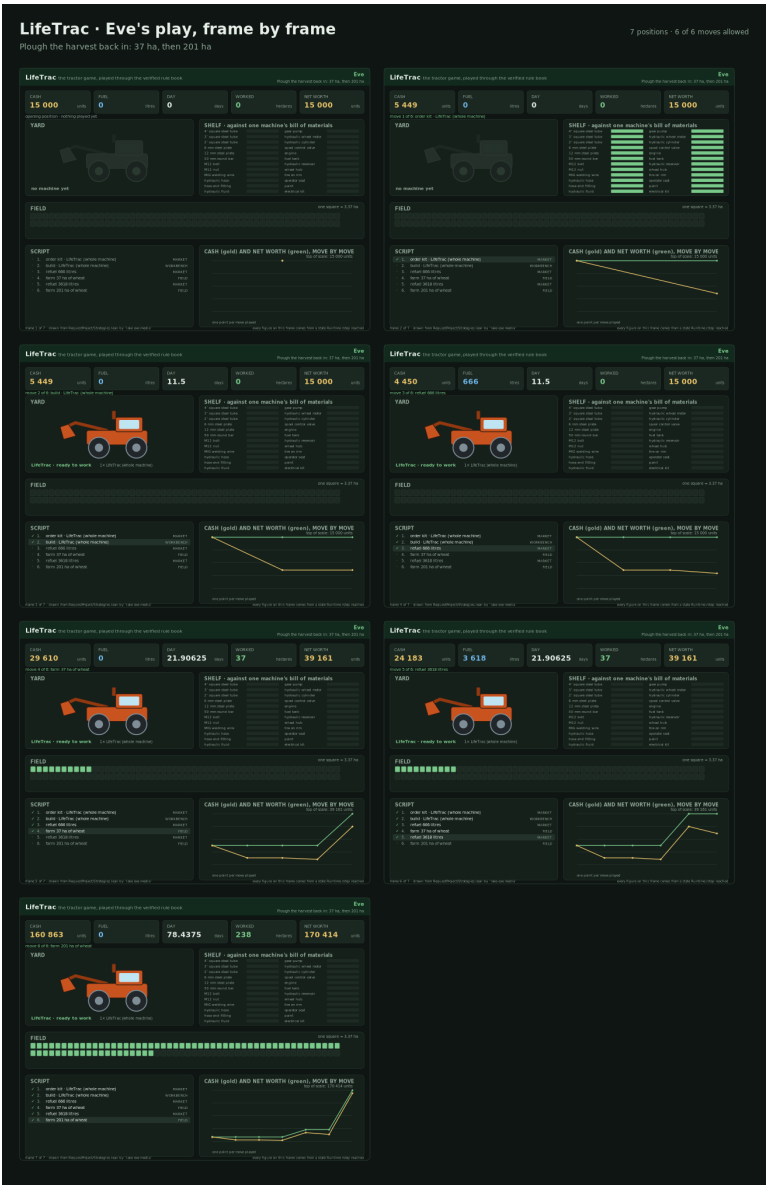
AND TWO WAYS TO FARM IT

Dan spends every unit he has on the first crop and puts 37 hectares in — one more is refused, and that refusal is proved too. Eve does what Dan does and then ploughs the harvest straight back into the ground: 238 hectares by day 78.4375, and 170 414 of net worth.

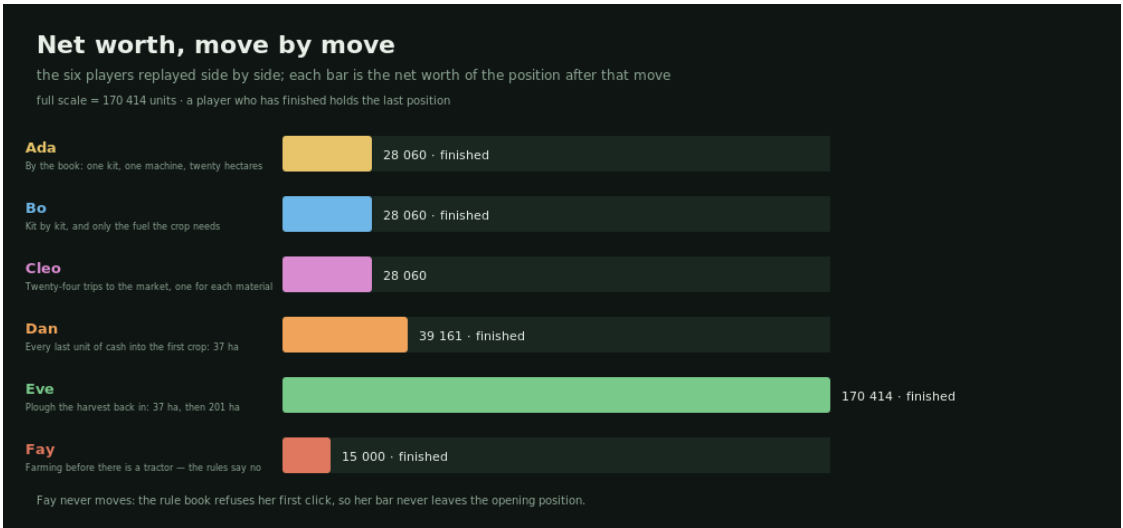


Fay clicks *farm 20 ha of wheat* on the first morning. There is no machine in the yard, so the rule book has no position to move to and the click does nothing: the move is refused, and the game proves it is (`fay_refused`).

A season, frame by frame



Eve's play in 7 positions: the opening, the kit on the shelf, the machine built, the tank filled, 37 hectares worked, the tank filled again out of the harvest, and 201 hectares more.



Net worth after every move. In the animated version of this picture the bars grow move by move; a still viewer shows the finish.

Why the pictures can be believed

Every frame is a function of a position the rule book produced. The renderer takes an `RState` — the state of the game in machine integers — and draws it; the states come from replaying a script through `rstep`, one move at a time, and the proof `traceStates_getLast` says the last state of the sequence the pictures are drawn from is the state the game itself ends on. There is no separate model of the game for the pictures to drift away from.

Theorem	What it says
<code>ada_plays, bo_plays, cleo_plays, dan_plays, eve_plays kits_equal_whole_kit</code>	Each script is legal from the first move to the last, and ends on exactly the cash, fuel, calendar, acreage and net worth printed here. Ordering the six sub-assembly kits puts exactly the whole machine's bill of materials on the shelf, and costs exactly the same.
<code>bo_netWorth_eq_ada</code>	Two different routes to the same machine end the season worth the same, to the unit.
<code>cleo_cash_add_fifteen</code>	Buying the materials one by one costs exactly 15 more, and nothing else differs.
<code>dan_maximal</code>	One more hectare in the first season is refused: after the machine and the fuel the seed is unaffordable.
<code>fay_refused, fay_stops_at_once</code>	Farming with no machine in the yard has no result, and the play stops at that move.
<code>scoreboard</code>	The order of the finish: Cleo, Ada, Bo, Dan, Eve.
<code>cash_nonneg_along_play</code>	No player is ever in debt, at any position of any of the six plays.

The whole development

The game sits on top of a Lean 4 formalization of the machine itself: the skid-steer kinematics and the proof that the forward and inverse maps are mutually inverse; the hydrostatic drive, where the mechanical power at the shaft is proved never to exceed the hydraulic power supplied; the tipping and sliding limits of a machine on a slope; the loader linkage and the cylinder that drives it; the production chain from ore to tractor, with lower bounds on what any bootstrap can cost; and the user interface, whose click log is proved to replay to the position on the screen. The rule book is compiled to WebAssembly, the compiler is proved correct against the rule book, and the emitted bytes are proved to decode back to the module the proofs are about.

PLAY IT

`lake exe emit web` writes `web/lifetrac.wasm` and `web/index.html`; open the page and play. `node web/test.mjs` replays the verified season through the emitted module.

MAKE THESE PICTURES AGAIN

`lake exe media media` writes the SVG — one frame per position, a storyboard per play, an animated SVG per play, and the two charts — and `python3 tools/rasterize.py media` writes the PNG and the animated PNG. `tectonic paper/brochure.tex` rebuilds this brochure.

LifeTrac is a machine of the Open Source Ecology Global Village Construction Set:

<https://github.com/OpenSourceEcology/LifeTrac>

The formalization, the game, the pictures and this brochure are in the same repository.